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Digital Care Plan for Sickle Cell

A partnership to make acute crisis care safer, faster and fairer

The NHS Race & Health Observatory is seeking one NHS acute trust to partner on the pilot of a personalised digital care plan for people with sickle cell disease during an acute painful crisis. A working prototype already exists. You can try it today.

The problem

People with sickle cell disease do not reliably receive timely pain relief during an acute painful crisis. NICE guidance is unambiguous: analgesia should be offered **within 30 minutes** of arrival and the patient reviewed every 30 minutes thereafter (NICE CG143), yet this standard is routinely missed. The 2021 All-Party Parliamentary Group inquiry *No One's Listening* documented **avoidable deaths and a pattern of substandard care**, with patients facing delays, having their pain disbelieved, and being treated without dignity.

This is, at once, a **patient-safety failure** and a **health inequality**: sickle cell disease overwhelmingly affects Black African and Caribbean communities, and the care gap is a measurable expression of structural racism in the system. Acting on it is core business for any board, under patient-safety and CQC duties, and under the health-inequalities duty in the Health and Care Act 2022.

What we are building and what already exists

In 2022–23 the Observatory, with University Hospitals Bristol and Weston, ran discovery research into acute sickle cell care. It found that **personalised digital care plans** show real promise for improving access to, experience of, and accountability of care, but have not been widely tested or adopted.

We are now commissioning the build of that service: a personalised care plan, with an at-a-glance **acute crisis view** a clinician can act on in seconds: usual analgesia and opioid tolerance, allergies, the patient's baseline observations, escalation, named consultant, pregnancy flag and emergency contacts. It is accessible to clinicians, patients and their nominated carers.

This is not a concept on paper. A functional prototype and a project website are already live and open source:

- **Prototype (try it):** <https://digital-sickle.carlreynolds.net>
- **Project website:** <https://carlreynolds.net/digital-sickle/>

That existing build substantially de-risks delivery and shows exactly what your teams would be piloting.

Why partner with us

- **Act on a known safety risk.** A clear, fast, personalised plan reduces delayed analgesia and medication error in one of the highest-risk acute presentations your ED handles.
- **Demonstrable action on health inequalities.** A concrete, board-level response to a documented failing: evidence for CQC, your health-inequalities duty, and your equality objectives.
- **Be the national first-mover.** The pilot trust will be the visible exemplar for a service intended to spread across the NHS.
- **No vendor lock-in.** The service is built in the open; the intellectual property is intended to be **public domain**, free for any trust to adopt. You help build a national asset, not a proprietary product.
- **Support your staff.** Clear, specialist-approved guidance is of particular help to emergency department teams who may rarely manage sickle crises.

How this works with London's Universal Care Plan (UCP)

London's Universal Care Plan is welcome and important: it proves the model: most sickle cell patients in London now have a digital care plan their urgent-care teams use. This programme is designed to **complement and interoperate with that work, not compete with it.** What makes it distinct holds true whatever UCP's future footprint:

- **Open, not proprietary.** Open-source with public-domain intellectual property, so any trust can adopt it and connect it to existing records, with no licensing or vendor lock-in.
- **Purpose-built for the acute crisis, co-designed with patients.** An at-a-glance crisis view built around the NICE 30-minute decision and patient dignity, not a module of a generalist tool.
- **Accountability by design.** A visible log of who accessed a plan and patient feedback to challenge substandard care, directly answering the failings in *No One's Listening*.

UCP is, today, a London programme funded by London's integrated care boards; most of England has no equivalent. **Where UCP operates, this connects to it; where it does not, it provides the whole service.** Appendix A sets this out in full.

The ask

We are seeking a strategic partner that delivers sickle cell and acute care to join us in an **initial phase that completes by 31 March 2027** and delivers a stand-alone output. The partner would provide:

- **Match funding of £200,000**, paid to the Observatory, which contracts for delivery; and
- **Institutional support** for the initial private beta, principally the engagement of clinical staff in **Haematology** and **Accident & Emergency**, and IT support for workflow integration.

We would begin with a **Memorandum of Understanding for this initial phase.** The Observatory leads

the programme, retains overall control, and conducts the procurement, commercial, technical and legal work. This is a deliberately **bounded, time-defined and low-risk** commitment: it delivers a **stand-alone, workable digital service** by 31 March 2027, so the work has lasting value whether or not it continues. Any continuation beyond that date would be subject to a further agreement and confirmed onward funding.

Timeline

We propose a defined initial phase that completes by **31 March 2027** and delivers a working digital service. Work beyond that date is subject to confirmed onward funding.

Phase	Period	Focus
Partnership and discovery	2026	Agree partnership, set-up, user research and design
Build and initial beta to 31 March 2027		Build and deploy a working digital service, and begin a private beta with selected patients
Continuation	beyond March 2027 (subject to funding)	Wider pilot, iteration and national spread

The proof

- Working **prototype** and **website** (links above), with all code open at github.com/drcjar/digital-sickle
- Observatory & Public Digital, *Sickle Cell Digital Discovery Report* (2023)
- *No One's Listening*, APPG on Sickle Cell & Thalassaemia (2021)
- Telfer et al., *Blood Reviews* (2024); Tanabe et al., *Am J Hematol* (2018)

Next step

We would welcome a **30-minute conversation** with you and your haematology and emergency-care leads to discuss partnership.

Dr Carl Reynolds, Clinical Advisor (carl.reynolds@nhsrho.org)

Professor Habib Naqvi MBE, Chief Executive, NHS Race & Health Observatory

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Appendix A: How this complements London's Universal Care Plan (UCP)

London's Universal Care Plan deserves real credit. Since sickle cell was added in March 2024 it has reached around **74% of the estimated 10,500 sickle cell patients in London** (more than 7,800 plans), and urgent-care use has risen sharply (about 3,400 views in a year, up from 570). UCP settles the question the Observatory's research first raised: **digital sickle cell care plans are adopted, used, and**

valued. We treat UCP as a partner and a proof point, not a rival, and this programme is deliberately designed to interoperate with it.

What this programme adds is distinct and, importantly, **holds true whatever UCP's footprint becomes:**

1. It is open, not proprietary. UCP is delivered on a commercial platform (Better), funded by London's five integrated care boards. This service is **open-source, with public-domain intellectual property**, built once and adoptable by any trust or region without licensing or vendor lock-in, and designed to connect to existing shared records rather than replace them. Even if UCP were to extend nationally, an open, freely-adoptable, interoperable sickle-cell layer remains valuable, and complementary.

2. It is purpose-built for the acute crisis and co-designed with patients. UCP began in 2022 as an end-of-life and palliative-care planning tool; sickle cell is one module among many (alongside dementia, frailty and palliative care), with the pain plan held inside a generic "symptom management" section. We are designing **specifically for the acute painful crisis and the NICE 30-minute decision:** an at-a-glance crisis view co-produced with sickle cell patients and emergency and haematology clinicians, around the dignity and needs of a young population living with a lifelong condition.

3. It builds accountability into the product. The failings exposed in *No One's Listening* were failings of accountability: patients dismissed and unheard. Beyond a shared clinical record, this service gives patients and their nominated carers a **visible log of who has accessed or changed their plan**, structured **post-encounter feedback**, and a co-produced, portable record they can use to challenge inconsistent or substandard care.

4. It is built to be evaluated. UCP reports adoption; this programme is designed to **test whether the service improves the 30-minute standard and patient experience**, generating the evidence of clinical and economic benefit that the literature currently finds inadequate (Telfer et al., 2024), and feeding national registries and dashboards.

5. It is national and open from the outset. UCP is, today, a London programme funded by London's ICBs (around 100,000 Londoners across all conditions; no national rollout announced, and out-of-London access via the National Record Locator is view-only: a patient's own GP cannot update it, nor can the patient via the NHS App). Most of England therefore has no equivalent. This programme is national and open by design: **where UCP operates, the two can interoperate and share open components; where it does not, this provides the whole service.** The aim is one all parties share: **that no patient's care depends on which city they fall ill in.**

Appendix B: What a well-designed digital care plan can do

A care plan co-produced with patients and clinical specialists has the potential to improve, and in places transform, sickle cell care.

Patient safety & clinical care

- Faster, personalised analgesia during a crisis, removing the need to re-narrate a complex history while in severe pain and a vulnerable state.
- Reduced risk of medication and treatment error, particularly for patients managing sickle cell alongside other conditions or with acquired adverse reactions.
- Formal documentation of treatment history, prior medication responses and opioid-tolerance thresholds, creating a clinical record that is harder to override through bias or assumption.

- Plans accessible to 111 / 999, so the pre-hospital response is also informed.
- Pregnancy and reproductive-health flags, linking obstetric and haematology teams and preventing contraindicated prescribing.
- The ability to capture avoidance behaviour and associated safety risk: patients not seeking care after previous traumatic or discriminatory encounters.

Care continuity & coordination

- A persistent, portable plan that travels with the patient, improving handover from paediatric to adult services, and supporting care in unfamiliar or non-specialist settings (including abroad).
- Reduced duplicate diagnostics when patients present away from their home base.
- Flags for unmet referral needs (pain specialists, psychology, social care, genetic counselling, obstetrics) that would otherwise go unaddressed between episodes.
- Prompts for timely review, reducing patients falling through gaps and enabling proactive outreach rather than reactive, crisis-only contact.

Accountability

- A documented, co-produced, portable record patients can use to challenge inconsistent or substandard treatment.
- A log of the clinicians and staff involved in each episode.
- Visibility of patterns of dismissal, delayed analgesia or non-adherence at individual, organisational and system level, enabling targeted intervention, training and performance measurement.
- Post-encounter patient feedback, generating patient-reported experience data for commissioners, policymakers and providers.
- Formal documentation of carer roles, preferences and emergency contacts, legitimising carer involvement.

Policy, practice & research

- More accurate needs assessment and capacity planning from real-world data on crisis frequency, duration and acuity.
- Identification of high-frequency attenders who may need better disease management or escalated community support.
- A feed into national registries, near-real-time dashboards and research datasets, highlighting variation in demand, provision, treatment availability and outcomes (by geography, socioeconomic status and demographics) to support commissioning, workforce planning and advocacy.